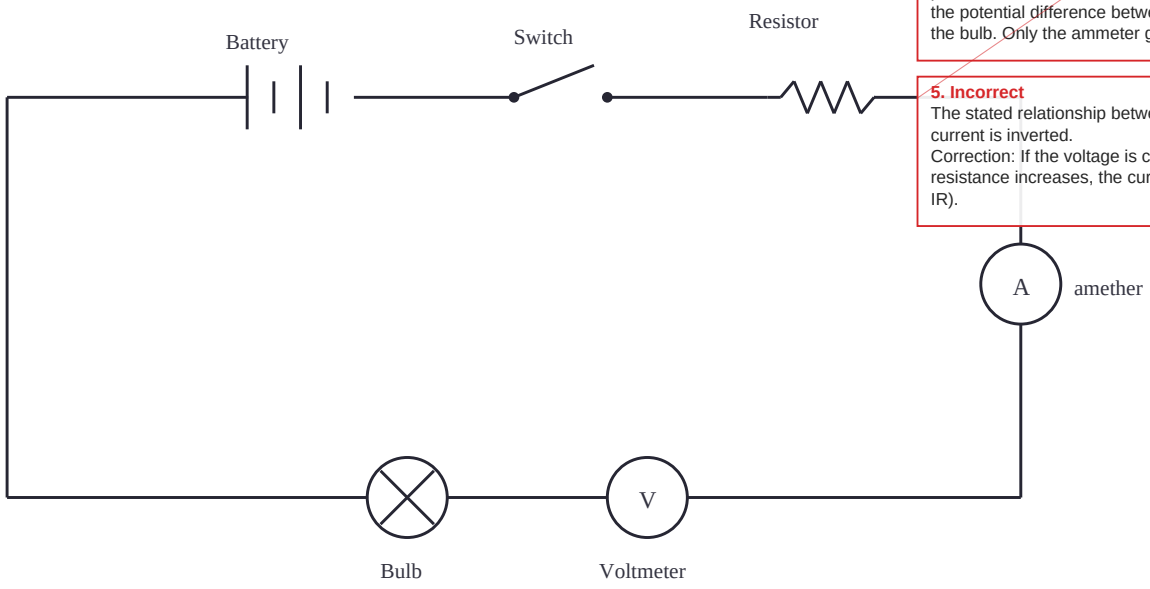
Name: Aarav Mehta Class: 10-B Subject: Combined Paper Total: 15 marks

Question 1

A simple electric circut is a closed path which allows curent to flow through it. The battery provides the difference which pushes the current around the loop. The switch is used to open and close the circuit. When the switch is closed the path is complete and the current flows through the bulb, and when the switch is open the path is broken so no curent can flow.

In my circut the battery, switch, resistor, bulb and the amether are all joined one after another in series in the main loop. The amether is put in series because it has to measure the curent that passes through the circuit. The voltmeter is also connected in series with the bulb so that it can measure the voltage of the bulb.

If the resistance of the circut is increased then the current also increases, because a bigger resistance means more current through the wire.



Question 2

Technology has changed the way studnets learn today. In my opinion technology makes learning better becuase information which was very hard to find earlier is now available in a few seconds on the internet. A student who does not understand a topic in the class can watch a video at home and understand properly. For example when I could not understand the water cycle in the class, I watched two videos on it and after that I understood the whole topic very well.

Technology also saves a lot of time becuase students do not have to search in many books in the library, they get all the informations immediately and they can spend that time on practice and revision instead which improves their knowlege and their marks in the exam also improve a lot and this is why every school should provide all the students and the government should also provide free internet to every student in the country.

Question 3

The table shows the demand and the supply of the product at different prices. I have drawn the demand and the suply curve on the graph below. I have taken the price on the horizontal axis and the quantity on the vertical axis. The demmand curve slopes downward because when the price is low the people are and the

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Correction: circuit

2. **Spelling**
Spelling: "curent" should be "current".
Correction: current

3. **Spelling**
Spelling: "amether" should be "ammeter".
Correction: ammeter

4. **Incorrect**
Substantive error: a voltmeter in series with the bulb would not measure the potential difference across it.
Correction: The voltmeter must be connected in parallel across the bulb, because it measures the potential difference between the two ends of the bulb. Only the ammeter goes in series.

5. **Incorrect**
The stated relationship between resistance and current is inverted.
Correction: If the voltage is constant and the resistance increases, the current decreases (V = IR).

6. **Spelling**
Spelling: "studnets" should be "students".
Correction: students

7. **Spelling**
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8. **Spelling**
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Correction: knowledge

9. **Spelling**
Spelling: "demmand" should be "demand".
Correction: demand

10. **Incorrect**
The axes are the wrong way round.
Correction: Quantity belongs on the horizontal axis and price on the vertical axis; the curves are drawn against those axes.

11. **Spelling**
Spelling: "suply" should be "supply".
Correction: supply

suply curve slopes upward because when the price is high the sellers want to sell more.

12 The two curves cut each other at the price of Rs 40 and the quantity of 40 units, and this po… the market.

When the price is below the equilibrium price the quantity demanded becomes more than t… there is a shortage in the market and the price starts rising. When the price is above the equilibrium price the quantity supplied becomes more than the quantity demanded, so there is a surplus in the market and the price starts falling.

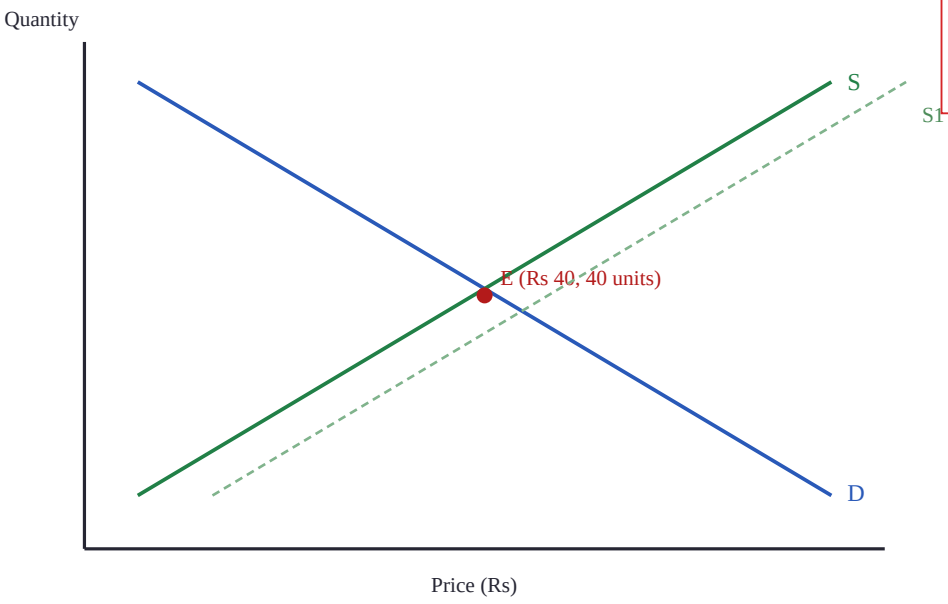
13 If the cost of production increases then the producer has to spend more money on making t…

14 supply curve shifts to the right side. Because of this the new equilibrium will be at a lower quantity than before.

12. Incorrect
The equilibrium is read off the schedule incorrectly.
Correction: Equilibrium is at a price of Rs 30 and a quantity of 60 units - the only row of the schedule where quantity demanded equals quantity supplied.

13. Incorrect
The direction of the supply shift is reversed.
Correction: A rise in production cost makes supplying less profitable at every price, so the supply curve shifts to the LEFT / upward.

14. Incorrect
The resulting movement of equilibrium price and quantity is stated backwards.
Correction: The new equilibrium is at a higher price and a lower quantity, assuming demand is unchanged.



Marking summary

6 / 15

Confidence 55% **NEEDS TEACHER REVIEW**

Graded by mock (offline-heuristic-v1) - degraded mode on 2/9/2026, 6:41:30 am

Q1 - Science - 2 / 5

[1/1] Correctly represents the main circuit with battery, switch, bulb and resistor connected in a closed series path

This rubric point is met.

Evidence: "In my circut the battery, switch, resistor, bulb and the amether are all joined one after another in series in the main loop."

[0/1] Correct placement of ammeter in series and voltmeter in parallel across the bulb

Substantive error: a voltmeter in series with the bulb would not measure the potential difference across it.

Evidence: "The voltmeter is also connected in series with the bulb so that it can measure the voltage of the bulb."

[1/1] Correct explanation of current flow and the function of the main components

This rubric point is met.

Evidence: "Question 1 A simple electric circut is a closed path which allows curent to flow through it."

[0/1] Correctly explains the relationship between resistance and current, including the relevant principle/Ohm's law

The stated relationship between resistance and current is inverted.

Evidence: "If the resistance of the circut is increased then the current also increases, because a bigger resistance pushes more current through the wire."

[0/1] Clear, logically structured explanation with appropriate labels and current direction in the diagram

The diagram lacks clear labelling or does not show the direction of conventional current.

Q2 - English - 3 / 5

[1/1] Presents a clear position on whether/how technology affects learning

This rubric point is met.

Evidence: "In my opinion technology makes learning much easier and better becuase information which was very hard to find earlier is now available in a few seconds on the internet."

[1/1] Provides relevant and logically developed arguments supporting the position

This rubric point is met.

Evidence: "In my opinion technology makes learning much easier and better becuase information which was very hard to find earlier is now available in a few seconds on the internet."

[0/1] Recognises and meaningfully addresses an opposing viewpoint or limitation

No opposing viewpoint or limitation is considered; the question explicitly asks for one.

[1/1] Uses relevant examples and demonstrates reasoning rather than merely making unsupported claims

This rubric point is met.

Evidence: "For example when I could not understand the water cycle in the class, I watched two videos on the internet and after that I understood the whole topic very well."

[0/1] Provides a coherent conclusion that follows from the discussion, with clear overall communication

There is no conclusion that follows from the discussion.

Q3 - Economics - 1 / 5

[0/1] Correctly plots and labels the demand and supply curves, with appropriate axes and direction

The axes are the wrong way round.

Evidence: "I have taken the price on the horizontal axis and the quantity on the vertical axis."

[0/1] Correctly identifies the equilibrium at Rs 30 and 60 units and explains why it is equilibrium

The equilibrium is read off the schedule incorrectly.

Evidence: "The two curves cut each other at the price of Rs 40 and the quantity of 40 units, and this point is the equilibrium of the market."

[1/1] Correctly explains shortage below equilibrium and surplus above equilibrium

This rubric point is met.

Evidence: "When the price is below the equilibrium price the quantity demanded becomes more than the quantity supplied, so there is a shortage in the market and the price starts rising."

[0/1] Correctly explains that increased production costs shift the supply curve left/upward

The direction of the supply shift is reversed.

Evidence: "If the cost of production increases then the producer has to spend more money on making the product, so the supply curve shifts to the right side."

[0/1] Correctly explains the resulting tendency toward a higher equilibrium price and lower equilibrium quantity, with the change represented appropriately on the graph

The resulting movement of equilibrium price and quantity is stated backwards.

Evidence: "Because of this the new equilibrium will be at a lower price and a higher quantity than before."

Review flags

- Graded without a language model, using keyword and pattern rules only.

Annotations on the paper

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Notes that could not be placed on the page

These relate to points the student did not write, or to quotes that could not be located in the paper.

- [Missing] The diagram lacks clear labelling or does not show the direction of conventional current.
- [Missing] No opposing viewpoint or limitation is considered; the question explicitly asks for one.
- [Missing] There is no conclusion that follows from the discussion.

Overall

Graded offline by the deterministic pattern grader (no model available). Marks are indicative and every rubric point should be confirmed by a teacher.